APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

IFRS 9 Expected Credit Loss Model

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Owner	Dr. Priya Nair (CRO)
Validator	Dr. Samuel Achebe (Model Validation Officer)
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Regulatory Framework: IFRS 9.5.5, EBA/GL/2020/06, SR 11-7

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1 Model Overview

1.1 Purpose

The IFRS 9 ECL model computes Expected Credit Loss provisions for the loan portfolio under the three-stage impairment framework. Provisions are recognised on a 12-month basis (Stage 1) or lifetime basis (Stage 2/3) depending on significant increases in credit risk (SICR).

1.2 Business Use

- **Financial reporting:** ECL provision recorded on income statement; affects CET1 ratio via retained earnings.
- **Credit risk management:** Stage classification drives credit officer workout strategy.
- **Capital planning:** Stressed ECL feeds DFAST adverse scenario P&L projections.
- **Pricing:** Risk-adjusted return on equity uses ECL as expected loss charge.

2 Theoretical Basis

2.1 ECL Formula

$$\text{ECL} = PD \times LGD \times EAD \times \text{DF}$$

For each obligor i across periods t:

$$\text{ECL}_i = \sum_{t=1}^{T_i} PD_i(t \mid t-1) \times LGD_i \times EAD_i(t) \times \text{DF}(t)$$

2.2 Stage Definitions

Stage 1 (no SICR): $\text{ECL}_{S1} = PD_{12m} \times LGD \times EAD$

Stage 2 (SICR): $\text{ECL}_{S2} = \sum_{t=1}^{T_{\text{life}}} PD_{\text{marg}}(t) \times LGD \times EAD(t) \times \text{DF}(t)$

Stage 3 (default, >90 DPD): $\text{ECL}_{S3} = LGD \times EAD_{\text{default}}$

2.3 SICR Definition

An obligor is reclassified Stage 1→2 if *any* of:

- Credit rating downgrade ≥ 2 notches since origination.
- Days past due > 30.
- Watch-list or watch-list-equivalent designation.
- Qualitative override (credit officer judgement).

2.4 PD Estimation

Through-the-cycle (TTC) PD from Moody's idealized cumulative default rates, adjusted via macro satellite model:

$$PD_{\text{PIT}}(t) = PD_{\text{TTC}} \times \exp\big(\beta_{\text{GDP}}\,\Delta\text{GDP}(t) + \beta_{\text{UR}}\,\Delta\text{UR}(t)\big)$$

$$\beta_{\text{GDP}} = -0.8 \text{ (GDP growth } +1\% \Rightarrow PD - 8\%), \quad \beta_{\text{UR}} = +0.6 \text{ (unemployment } +1\text{pp} \Rightarrow PD + 6\%).$$

2.5 LGD and EAD

LGD:

$$LGD = 1 - \text{Recovery}, \qquad \text{Recovery} = \min\left(\frac{\text{Collateral}}{EAD},\ 0.90\right) \times \text{haircut}$$

Unsecured: $LGD = 0.45$ (corporate), 0.25 (secured mortgages). Haircuts: CRE 30%, residential 10%.

EAD: $EAD = \text{drawn} + CCF \times \text{undrawn}$, where $CCF = 0.75$ (revolving), 0.50 (term).

3 Mathematical Specification

Sample portfolio: 50 obligors.

Parameter	Value
Rating distribution	60% IG (BBB+/BBB/BBB−), 30% BB, 10% B/CCC
Average PD (Stage 1)	1.2%
Average LGD	42%
ECL coverage (normal)	∼1.8%
ECL coverage (stressed)	∼3.5%

4 Implementation

Code location: `infrastructure/credit/ifrs9_ecl.py`
Class: `IFRS9ECLEngine`

Key methods:

- `calculate_portfolio_ecl()` — full portfolio ECL.
- `classify_stages()` — SICR assessment.
- `run_scenario(macro_params)` — stressed ECL for DFAST.

Runtime: ∼5 ms for 50 obligors.

5 Validation

Stage classification: Validated against sample obligors with known PD migration history; accuracy 94%.

Macro sensitivity: β_{GDP} validated against EBA stress test results for comparable European banks; within ±15%.

Coverage ratio: 1.8% (normal) vs EBA peer average 1.6%; slightly conservative, acceptable.

6 Model Limitations

1. **TTC PD basis:** TTC PD underestimates current default risk in downturns; PIT adjustment partially corrects this.
2. **50-obligor sample:** Production portfolio has 850 obligors; demo portfolio is illustrative only.
3. **Flat LGD:** Collateral values not mark-to-market; static LGD may understate loss in collateral price declines.
4. **Macro satellite not post-COVID calibrated:** GDP/unemployment coefficients estimated pre-2020.

7 Open Findings

ID	Severity	Description	Status
ECL-F1	**Major**	Macro satellite model GDP coefficient ($\beta_{\text{GDP}} = -0.8$) not recalibrated post-COVID; EBA/GL/2020/06 requires calibration to reflect current economic uncertainty.	Open